

Healthcare Business Question	MS SQL Server Query	Output																												
Demographics and Diagnoses: What is the demographic profile of the patient population, including age and gender distribution?	<pre>SELECT gender, CASE WHEN DATEDIFF(year, date_of_birth, GETDATE()) BETWEEN 0 AND 17 THEN 'Pediatric' WHEN DATEDIFF(year, date_of_birth, GETDATE()) BETWEEN 18 AND 64 THEN 'Adult' ELSE 'Senior' END AS age_group, COUNT(*) AS patient_count FROM [Healthcare_Database].[dbo].Patients GROUP BY gender, CASE WHEN DATEDIFF(year, date_of_birth, GETDATE()) BETWEEN 0 AND 17 THEN 'Pediatric' WHEN DATEDIFF(year, date_of_birth, GETDATE()) BETWEEN 18 AND 64 THEN 'Adult' ELSE 'Senior' END ORDER BY patient_count Desc</pre>	<i>Note: If there are more than 20 rows, only the first 20 are shown.</i> <table><tr><th></th><th>gender</th><th>age_group</th><th>patient_count</th></tr><tr><td>1</td><td>Female</td><td>Adult</td><td>44</td></tr><tr><td>2</td><td>Male</td><td>Adult</td><td>26</td></tr><tr><td>3</td><td>Male</td><td>Senior</td><td>14</td></tr><tr><td>4</td><td>Female</td><td>Senior</td><td>7</td></tr><tr><td>5</td><td>Female</td><td>Pediatric</td><td>6</td></tr><tr><td>6</td><td>Male</td><td>Pediatric</td><td>3</td></tr></table>		gender	age_group	patient_count	1	Female	Adult	44	2	Male	Adult	26	3	Male	Senior	14	4	Female	Senior	7	5	Female	Pediatric	6	6	Male	Pediatric	3
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Demographics and Diagnoses: How do diagnoses vary across different demographic groups, including gender and age?	<pre>SELECT gender, diagnosis, CASE WHEN DATEDIFF(year, date_of_birth, GETDATE()) BETWEEN 0 AND 17 THEN 'Pediatric' WHEN DATEDIFF(year, date_of_birth, GETDATE()) BETWEEN 18 AND 64 THEN 'Adult' ELSE 'Senior' END AS age_group, COUNT(*) AS patient_count FROM [Healthcare_Database].[dbo].Patients AS p INNER JOIN [Healthcare_Database].[dbo].[Outpatient Visits] AS ov ON p.patient_id = ov.patient_id WHERE diagnosis <> 'Unknown' GROUP BY gender, diagnosis, CASE WHEN DATEDIFF(year, date_of_birth, GETDATE()) BETWEEN 0 AND 17 THEN 'Pediatric' WHEN DATEDIFF(year, date_of_birth, GETDATE()) BETWEEN 18 AND 64 THEN 'Adult' ELSE 'Senior' END ORDER BY patient_count DESC, diagnosis ASC</pre>	<table><tr><th></th><th>gender</th><th>diagnosis</th><th>age_group</th><th>patient_count</th></tr><tr><td>1</td><td>Female</td><td>Diabetes</td><td>Adult</td><td>20</td></tr><tr><td>2</td><td>Female</td><td>Hypertension</td><td>Adult</td><td>13</td></tr><tr><td>3</td><td>Female</td><td>Muscle Injury</td><td>Adult</td><td>12</td></tr><tr><td>4</td><td>Female</td><td>Respiratory Illness</td><td>Adult</td><td>12</td></tr><tr><td>5</td><td>Male</td><td>Diabetes</td><td>Adult</td><td>11</td></tr><tr><td>6</td><td>Male</td><td>Hypertension</td><td>Adult</td><td>9</td></tr><tr><td>7</td><td>Female</td><td>Hyperlipidemia</td><td>Adult</td><td>7</td></tr><tr><td>8</td><td>Male</td><td>Muscle Injury</td><td>Adult</td><td>6</td></tr><tr><td>9</td><td>Male</td><td>Respiratory Illness</td><td>Senior</td><td>6</td></tr><tr><td>10</td><td>Female</td><td>Migraine</td><td>Adult</td><td>5</td></tr><tr><td>11</td><td>Male</td><td>Allergic Reaction</td><td>Adult</td><td>4</td></tr><tr><td>12</td><td>Male</td><td>Common Cold</td><td>Adult</td><td>4</td></tr><tr><td>13</td><td>Male</td><td>Diabetes</td><td>Senior</td><td>4</td></tr><tr><td>14</td><td>Female</td><td>Ear Infection</td><td>Pediatric</td><td>4</td></tr><tr><td>15</td><td>Male</td><td>Muscle Injury</td><td>Senior</td><td>4</td></tr><tr><td>16</td><td>Male</td><td>Respiratory Illness</td><td>Adult</td><td>4</td></tr><tr><td>17</td><td>Female</td><td>Respiratory Illness</td><td>Senior</td><td>4</td></tr><tr><td>18</td><td>Male</td><td>Bronchiolitis</td><td>Pediatric</td><td>3</td></tr><tr><td>19</td><td>Female</td><td>Bronchiolitis</td><td>Pediatric</td><td>3</td></tr><tr><td>20</td><td>Female</td><td>Common Cold</td><td>Adult</td><td>3</td></tr></table>		gender	diagnosis	age_group	patient_count	1	Female	Diabetes	Adult	20	2	Female	Hypertension	Adult	13	3	Female	Muscle Injury	Adult	12	4	Female	Respiratory Illness	Adult	12	5	Male	Diabetes	Adult	11	6	Male	Hypertension	Adult	9	7	Female	Hyperlipidemia	Adult	7	8	Male	Muscle Injury	Adult	6	9	Male	Respiratory Illness	Senior	6	10	Female	Migraine	Adult	5	11	Male	Allergic Reaction	Adult	4	12	Male	Common Cold	Adult	4	13	Male	Diabetes	Senior	4	14	Female	Ear Infection	Pediatric	4	15	Male	Muscle Injury	Senior	4	16	Male	Respiratory Illness	Adult	4	17	Female	Respiratory Illness	Senior	4	18	Male	Bronchiolitis	Pediatric	3	19	Female	Bronchiolitis	Pediatric	3	20	Female	Common Cold	Adult	3
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Demographics and Diagnoses: Which diagnoses are more prevalent among patients?	<pre>SELECT TOP 5 diagnosis, COUNT(*) AS total_patient_count FROM [Healthcare_Database].[dbo].[Outpatient Visits] WHERE diagnosis <> 'Unknown' GROUP BY diagnosis ORDER BY total_patient_count DESC</pre>	<table><tr><th></th><th>diagnosis</th><th>total_patient_count</th></tr><tr><td>1</td><td>Diabetes</td><td>37</td></tr><tr><td>2</td><td>Respiratory Illness</td><td>26</td></tr><tr><td>3</td><td>Muscle Injury</td><td>24</td></tr><tr><td>4</td><td>Hypertension</td><td>23</td></tr><tr><td>5</td><td>Common Cold</td><td>13</td></tr></table>		diagnosis	total_patient_count	1	Diabetes	37	2	Respiratory Illness	26	3	Muscle Injury	24	4	Hypertension	23	5	Common Cold	13																																																																																							
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Appointments: What are the most common appointment times in the day, and	<pre>SELECT DATEPART (hour, appointment_time) AS appointment_hour, COUNT(*) AS appointment_count FROM [Healthcare_Database].[dbo].Appointments GROUP BY DATEPART (hour, appointment_time) ORDER BY appointment_count DESC</pre>	<table><tr><th></th><th>appointment_hour</th><th>appointment_count</th></tr><tr><td>1</td><td>12</td><td>46</td></tr><tr><td>2</td><td>9</td><td>40</td></tr><tr><td>3</td><td>11</td><td>38</td></tr><tr><td>4</td><td>10</td><td>34</td></tr><tr><td>5</td><td>8</td><td>16</td></tr></table>		appointment_hour	appointment_count	1	12	46	2	9	40	3	11	38	4	10	34	5	8	16																																																																																							
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how does the distribution vary?																																																																																						
Labs: What are the most commonly ordered lab tests?	<pre>SELECT test_name, COUNT(*) AS test_count FROM [Healthcare_Database].[dbo].[Lab Results] GROUP BY test_name ORDER BY test_count DESC</pre>	<table><tr><th></th><th>test_name</th><th>test_count</th></tr><tr><td>1</td><td>Chloride</td><td>49</td></tr><tr><td>2</td><td>Fasting Blood Sugar</td><td>42</td></tr><tr><td>3</td><td>ALT</td><td>40</td></tr><tr><td>4</td><td>Thyroid Stimulating Hormone</td><td>40</td></tr><tr><td>5</td><td>White Blood Cells</td><td>39</td></tr><tr><td>6</td><td>Uric Acid</td><td>37</td></tr><tr><td>7</td><td>Hemoglobin A1C</td><td>35</td></tr><tr><td>8</td><td>C-Reactive Protein</td><td>35</td></tr><tr><td>9</td><td>HDL Cholesterol</td><td>34</td></tr><tr><td>10</td><td>Creatinine</td><td>29</td></tr></table>		test_name	test_count	1	Chloride	49	2	Fasting Blood Sugar	42	3	ALT	40	4	Thyroid Stimulating Hormone	40	5	White Blood Cells	39	6	Uric Acid	37	7	Hemoglobin A1C	35	8	C-Reactive Protein	35	9	HDL Cholesterol	34	10	Creatinine	29																																																			
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Labs: Typically, fasting blood sugar levels falls between 70-100 mg/dL. Our goal is to identify patients whose lab results are outside this normal range to implement early intervention.	<pre>SELECT p.patient_id, p.patient_name, result_value FROM [Healthcare_Database].[dbo].Patients AS p INNER JOIN [Healthcare_Database].[dbo].[Outpatient Visits] AS ov ON p.patient_id = ov.patient_id INNER JOIN [Healthcare_Database].[dbo].[Lab Results] AS lr ON ov.visit_id = lr.visit_id WHERE lr.test_name = 'Fasting Blood Sugar' AND (lr.result_value < 70 OR lr.result_value >100)</pre>	<table><tr><th></th><th>patient_id</th><th>patient_name</th><th>result_value</th></tr><tr><td>1</td><td>521001</td><td>Emma Johnson</td><td>107</td></tr><tr><td>2</td><td>521001</td><td>Emma Johnson</td><td>101</td></tr><tr><td>3</td><td>521002</td><td>Michael Smith</td><td>103</td></tr><tr><td>4</td><td>521004</td><td>William Jones</td><td>110</td></tr><tr><td>5</td><td>521004</td><td>William Jones</td><td>105.82</td></tr><tr><td>6</td><td>521008</td><td>Bethany Clark</td><td>58.95</td></tr><tr><td>7</td><td>521009</td><td>Jose Gonzalez</td><td>110</td></tr><tr><td>8</td><td>521011</td><td>Amelia Thomas</td><td>111.47</td></tr><tr><td>9</td><td>521011</td><td>Amelia Thomas</td><td>112</td></tr><tr><td>10</td><td>521013</td><td>Alexander Perez</td><td>55.87</td></tr><tr><td>11</td><td>521021</td><td>Olivia Scott</td><td>107.99</td></tr><tr><td>12</td><td>521024</td><td>Liam Baker</td><td>124.81</td></tr><tr><td>13</td><td>521030</td><td>James Cooper</td><td>66.38</td></tr><tr><td>14</td><td>521031</td><td>Andy Morris</td><td>127</td></tr><tr><td>15</td><td>521032</td><td>Rick Sanders</td><td>50.91</td></tr><tr><td>16</td><td>521033</td><td>Eva Torres</td><td>123.27</td></tr><tr><td>17</td><td>521047</td><td>Nick Reid</td><td>55.34</td></tr><tr><td>18</td><td>521069</td><td>Sara Martin</td><td>120</td></tr><tr><td>19</td><td>521074</td><td>Sophie Myers</td><td>105</td></tr><tr><td>20</td><td>521092</td><td>Ava White</td><td>110</td></tr></table>		patient_id	patient_name	result_value	1	521001	Emma Johnson	107	2	521001	Emma Johnson	101	3	521002	Michael Smith	103	4	521004	William Jones	110	5	521004	William Jones	105.82	6	521008	Bethany Clark	58.95	7	521009	Jose Gonzalez	110	8	521011	Amelia Thomas	111.47	9	521011	Amelia Thomas	112	10	521013	Alexander Perez	55.87	11	521021	Olivia Scott	107.99	12	521024	Liam Baker	124.81	13	521030	James Cooper	66.38	14	521031	Andy Morris	127	15	521032	Rick Sanders	50.91	16	521033	Eva Torres	123.27	17	521047	Nick Reid	55.34	18	521069	Sara Martin	120	19	521074	Sophie Myers	105	20	521092	Ava White	110
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Risk Stratification:

How many patients are considered High, Medium, and Low Risk?

High Risk: patients who are smokers and have been diagnosed with either hypertension or diabetes.

Medium Risk: patients who are non-smokers and have been diagnosed with either hypertension or diabetes.

Low Risk: patients who do not fall into the above categories.

```
SELECT
CASE
WHEN smoker_status = 'Y' AND (diagnosis = 'Hypertension' OR diagnosis = 'Diabetes') THEN 'High Risk'
WHEN smoker_status = 'N' AND (diagnosis = 'Hypertension' OR diagnosis = 'Diabetes') THEN 'Medium Risk'
ELSE 'Low Risk'
END AS Risk_category,
COUNT(patient_id) AS num_patients
FROM [Healthcare_Database].[dbo].[Outpatient Visits]
GROUP BY
CASE
WHEN smoker_status = 'Y' AND (diagnosis = 'Hypertension' OR diagnosis = 'Diabetes') THEN 'High Risk'
WHEN smoker_status = 'N' AND (diagnosis = 'Hypertension' OR diagnosis = 'Diabetes') THEN 'Medium Risk'
ELSE 'Low Risk'
END
```

	Risk_category	num_patients
1	High Risk	21
2	Low Risk	389
3	Medium Risk	39

Readmissions:

Which patients had multiple visits within 30 days of their previous medical visit?

```
SELECT
ov_initial.patient_id,
ov_initial.visit_date AS initial_visit_date,
ov_initial.reason_for_visit AS reason_for_initial_visit,
ov_readmit.visit_date AS readmission_date,
ov_readmit.reason_for_visit AS reason_for_readmission,
DATEDIFF(day, ov_initial.visit_date, ov_readmit.visit_date) AS days_between
FROM
[Healthcare_database].[dbo].[Outpatient Visits] AS ov_initial
INNER JOIN [Healthcare_database].[dbo].[Outpatient Visits] AS ov_readmit
ON ov_initial.patient_id = ov_readmit.patient_id
WHERE DATEDIFF(day, ov_initial.visit_date, ov_readmit.visit_date) <= 30
AND ov_readmit.visit_date > ov_initial.visit_date
```

See below for detailed output.

Detailed Output

	patient_id	initial_visit_date	reason_for_initial_visit	readmission_date	reason_for_readmission	days_between
1	521025	2024-12-07 00:00:00.000	Checkup	2024-12-31 00:00:00.000	Checkup	24
2	521026	2024-05-31 00:00:00.000	Knee pain	2024-06-10 00:00:00.000	Checkup	10
3	521027	2024-10-10 00:00:00.000	Diabetes check	2024-11-08 00:00:00.000	Annual physical	29
4	521027	2024-11-08 00:00:00.000	Annual physical	2024-11-10 00:00:00.000	Diet and Exercise Counseling	2
5	521028	2023-05-29 00:00:00.000	Fever	2023-06-15 00:00:00.000	Annual physical	17
6	521029	2024-01-19 00:00:00.000	Annual physical	2024-01-30 00:00:00.000	Diet and Exercise Counseling	11
7	521029	2024-01-19 00:00:00.000	Annual physical	2024-02-01 00:00:00.000	Diabetes check	13
8	521029	2024-01-30 00:00:00.000	Diet and Exercise Counseling	2024-02-01 00:00:00.000	Diabetes check	2
9	521030	2023-02-11 00:00:00.000	Checkup	2023-02-25 00:00:00.000	Checkup	14
10	521030	2024-10-30 00:00:00.000	Annual physical	2024-11-14 00:00:00.000	Diabetes check	15
11	521032	2024-03-10 00:00:00.000	Fever	2024-03-15 00:00:00.000	Headache	5
12	521032	2024-03-10 00:00:00.000	Fever	2024-03-30 00:00:00.000	Back pain	20
13	521032	2024-03-15 00:00:00.000	Headache	2024-03-30 00:00:00.000	Back pain	15
14	521033	2024-08-08 00:00:00.000	Knee pain	2024-08-27 00:00:00.000	Fever	19
15	521034	2024-08-01 00:00:00.000	Diabetes check	2024-08-16 00:00:00.000	Annual physical	15
16	521034	2024-11-10 00:00:00.000	Checkup	2024-11-23 00:00:00.000	Checkup	13
17	521034	2024-11-10 00:00:00.000	Checkup	2024-12-07 00:00:00.000	Back pain	27
18	521034	2024-11-23 00:00:00.000	Checkup	2024-12-07 00:00:00.000	Back pain	14
19	521035	2023-04-20 00:00:00.000	Back pain	2023-05-17 00:00:00.000	Hypertension	27
20	521035	2023-07-02 00:00:00.000	Headache	2023-07-23 00:00:00.000	Fever	21